Dev Chauhan

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Profile

click to view portfolio

Computer Science Engineering student at Nirma University (Third year, B.Tech, Class of 2028) with hands-on experience designing, building, and testing full-stack, AI-integrated systems and automated QA frameworks — from self-healing test automation to ML-driven monitoring pipelines. Strong foundation in data structures, algorithms, and object-oriented design, with a track record of translating ambiguous requirements into scalable, well-tested, production-style solutions across test automation, embedded systems, and distributed AI. Proven collaborator across hackathons and academic research, with a paper selected for publication at RIACT 2026. Checkout my Portfolio for an immersive experience

Education

Nirma University Ahmedabad, Gujarat
B.Tech in Computer Science and Engineering — Pursuing *Aug 2024 – July 2028*

Best High School Ahmedabad, Gujarat
Higher Secondary Certificate (12th Grade), 83% — Science & Mathematics *2009 – 2024*

Experience

Team Arrow (Robotics Club) Nirma University
Technical Team Member *Aug 2024 – Nov 2024*

- Engineered autonomous navigation software for a distributed, drone-based robotics platform, integrating the Ardupilot flight stack with custom communication modules for real-time autonomous control.
- Built, tested, and debugged embedded computer-vision and sensor pipelines on Jetson Nano and Raspberry Pi (Pi HQ Cameras), performing systematic hardware-software integration testing and root-cause debugging across the stack.

Projects & Publication

AI-Powered Self-Healing Test Automation Framework Personal Project

- Architected a self-healing Selenium/Python test automation framework that detects broken element locators, parses the live DOM, and auto-recovers via a rule-based similarity engine (tag, class, text, and attribute matching), eliminating brittle UI-driven test failures.
- Designed an element fingerprinting system capturing ID, class, text, tag, and XPath signatures, enabling accurate identification of replacement locators after front-end changes with up to ~92% match confidence.
- Built a structured self-healing event logger capturing failure diagnostics, candidate scoring, and confidence metrics for full test-run auditability and regression traceability.

AI/ML-Enhanced Sensor Calibration Platform Hackathon MVP (36 Hour)

- Designed and deployed an end-to-end, production-style calibration system with automated validation pipelines, reducing sensor measurement error by ~30% through automated offset-correction and noise-filtering algorithms.
- Built and rigorously tested anomaly-detection and drift-prediction models (Isolation Forest, LSTM, CNN-LSTM), achieving >95% detection accuracy via a SQL-backed (SQLite) pipeline and automated regression-reporting dashboard, cutting manual verification effort by ~40%.

VeriPhi | Browser-Integrated AI Trust Intelligence Layer Hackathon MVP

- Architected a Chrome extension (Manifest V3) that verifies LLM output correctness in real time, decomposing streaming text into atomic claims via NLP-based sentence-boundary detection for automated fact-checking.
- Designed a hybrid edge-cloud AI architecture (on-device Gemini Nano, cloud Gemini 1.5 Flash) with Vertex AI Grounding to cross-reference claims against live search data, validated for low-latency, privacy-preserving, reliable inference.

Research Paper — Autonomous Disaster Management Robot (International) RIACT 2026

- Designed a fully autonomous robotic system, published as "Development and Application of a Fully Autonomous Robot for Disaster Management and Humanitarian Purposes," to navigate hazardous, post-disaster environments and deliver humanitarian aid without human intervention.
- Engineered fail-safe navigation logic to eliminate physical risk exposure for human rescuers operating in unknown environments.

Skills

Programming Languages: C, C++, Python, JavaScript, Java, Node, React, Tailwind, MATLAB, MongoDB, SQL/NoSQL.

QA & Test Automation: Selenium WebDriver, Self-Healing Test Frameworks, Pytest, API Testing (Postman), Regression & Functional Testing, Bug Tracking (Jira), Git/GitHub, CI/CD (GitHub Actions), Agile/Scrum.

Software Engineering & ML: Machine Learning (Isolation Forest, LSTM, CNN-LSTM), NLP, SQL & Data Pipelines (SQLite), Chrome Extension Development (Manifest V3).

Embedded Systems & IoT: Arduino, Raspberry Pi, Jetson Nano, Embedded C, GSM Modules, Sensor Interfacing.

Co-Curricular & Certifications

- NASA Space Apps Challenge '24: Participant in the global hackathon solving space-related challenges.
- IoT Gas Leakage Detector: Engineered an offline MQ-6/GSM-based alert system delivering real-time SMS notifications without Wi-Fi dependency.